POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 4012 | Semester IV | Theory | 4 Credits

Theory of Structures

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 4012 as Theory of Structures in Semester IV for Civil Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses reliable public structural-analysis sources and is not represented as recovered official SITTTR Course 4012 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. Formulae and sign conventions must be matched to the course/instructor-approved convention; real design requires applicable standards and qualified professional responsibility.

Verified source note: The official detailed source was inaccessible. Public sources supporting beams, shear/moment, stresses, deflections, trusses and determinacy are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Theory of Structures develops the modelling and analysis skills used to determine reactions, internal forces, moments, stresses, deflections and stability of beams, trusses and frames. Clear free-body diagrams, consistent signs and equilibrium are the basis of every solution.

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Prerequisite foundation

Engineering mechanics, basic algebra, units, vectors and strength-of-materials fundamentals.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Model structural systems, loads and supports using correct free-body diagrams.
2. Determine reactions and draw/interpret shear-force and bending-moment diagrams.
3. Analyse simple trusses and beam stresses/deflections using appropriate methods.
4. Distinguish determinate from indeterminate systems and explain stability/buckling concepts.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Model loads, supports and reactions for elementary structural systems.	Apply
CO2	Determine and interpret shear-force and bending-moment diagrams for simple beams.	Apply
CO3	Analyse basic trusses, stresses and deflections using a stated method and assumptions.	Apply
CO4	Explain determinacy, compatibility, stability and column-buckling concepts.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Structural Modelling and Equilibrium	Types of structures, loads, supports, free-body diagrams, reactions, equilibrium, determinacy and sign conventions.	Approx. 14 hours	CO1	Module 1
2	Shear Force, Bending Moment and Stresses	Section cuts, shear/moment sign convention, SFD/BMD, load-shear-moment relations, bending and shear stress.	Approx. 16 hours	CO2	Module 2
3	Trusses and Deflections	Pin-jointed trusses, methods of joints/sections, zero-force members, strain energy/virtual work concepts, slope and beam deflection.	Approx. 16 hours	CO3	Module 3
4	Indeterminacy, Stability and Application	Compatibility, stiffness/flexibility overview, continuous beams/frames, columns/buckling, influence of loads and responsible use of analysis tools.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Structural Modelling and Equilibrium

Types of structures, loads, supports, free-body diagrams, reactions, equilibrium, determinacy and sign conventions.

CO1 · Approx. 14 hours

Shear Force, Bending Moment and Stresses

Section cuts, shear/moment sign convention, SFD/BMD, load-shear-moment relations, bending and shear stress.

CO2 · Approx. 16 hours

Trusses and Deflections

Pin-jointed trusses, methods of joints/sections, zero-force members, strain energy/virtual work concepts, slope and beam deflection.

CO3 · Approx. 16 hours

Indeterminacy, Stability and Application

Compatibility, stiffness/flexibility overview, continuous beams/frames, columns/buckling, influence of loads and responsible use of analysis tools.

CO4 · Approx. 14 hours

MODULE 1

Structural Modelling and Equilibrium

Types of structures, loads, supports, free-body diagrams, reactions, equilibrium, determinacy and sign conventions.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

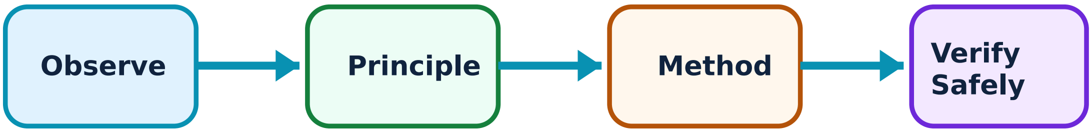
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Structural Modelling and Equilibrium: identify the system, state its principle, follow the correct method and verify the result safely.

1. Loads, supports and free-body diagrams−

A structural model represents members, supports and applied loads with idealised symbols. Pinned, roller and fixed supports restrain different motions and therefore develop different reactions.

Study and application method

Isolate the body, replace supports with unknown reactions, show all loads/dimensions and state the coordinate/sign convention before writing equations.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Isolate the body, replace supports with unknown reactions, show all loads/dimensions and state the coordinate/sign convention before writing equations.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never calculate reactions from an incomplete free-body diagram or without units.

2. Equilibrium and reactions−

A planar structure in static equilibrium satisfies zero resultant force in two directions and zero resultant moment. These equations determine reactions for statically determinate systems.

Study and application method

Choose moment centre strategically, solve equations systematically and check force/moment balance after obtaining results.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Keep distributed-load resultants, locations and signs correct; a correct magnitude at a wrong location gives a wrong moment.

3. Determinacy and stability−

Determinate structures can be solved using equilibrium alone; indeterminate structures need compatibility and stiffness/deformation relations. A model must also be stable against rigid-body motion.

Study and application method

Count unknown reactions/member forces relative to independent equations and inspect whether the support/member arrangement prevents translation and rotation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Do not confuse an over-restrained diagram with a stable, well-modelled real structure.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Shear Force, Bending Moment and Stresses

Section cuts, shear/moment sign convention, SFD/BMD, load-shear-moment relations, bending and shear stress.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

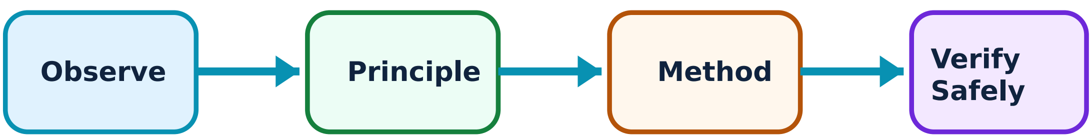
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Shear Force, Bending Moment and Stresses: identify the system, state its principle, follow the correct method and verify the result safely.

1. Section-cut internal actions−

A cut through a loaded member reveals internal axial force, shear force and bending moment needed to keep the isolated segment in equilibrium.

Study and application method

Find reactions first, cut at a location, choose one side, apply the declared sign convention and write equilibrium equations.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Find reactions first, cut at a location, choose one side, apply the declared sign convention and write equilibrium equations.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Use one sign convention consistently throughout diagrams and calculations.

2. Shear-force and bending-moment diagrams−

SFD/BMD show how internal shear and moment vary along a beam. Point loads create shear jumps; distributed loads change shear slope; extrema of moment occur where shear crosses zero.

Study and application method

Mark key positions, calculate values on either side of discontinuities, draw the correct line/curve shape and verify end/support conditions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Do not draw diagrams from memorised shapes before computing reactions and load intervals.

3. Bending and shear stress−

Bending moment produces normal stress varying with distance from the neutral axis; transverse shear produces a non-uniform shear-stress distribution related to section geometry.

Study and application method

State assumptions, identify M/V and section properties, use consistent units and locate the fibre/point where stress is required.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State assumptions, identify M/V and section properties, use consistent units and locate the fibre/point where stress is required.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not apply a beam formula outside its assumptions or mix units for force, length and section properties.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Trusses and Deflections

Pin-jointed trusses, methods of joints/sections, zero-force members, strain energy/virtual work concepts, slope and beam deflection.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

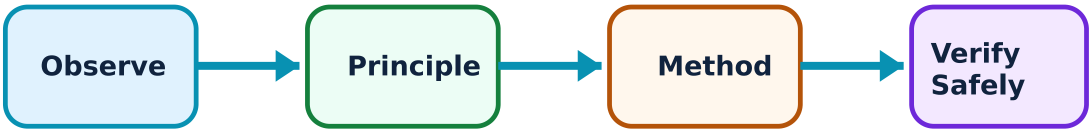
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Trusses and Deflections: identify the system, state its principle, follow the correct method and verify the result safely.

1. Pin-jointed truss analysis−

An ideal plane truss carries loads at joints and members act mainly in axial tension or compression. Method of joints uses joint equilibrium; method of sections isolates a portion efficiently.

Study and application method

Check global reactions, label members/joints, apply method of joints or a strategic section and state tension/compression sign convention.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Check global reactions, label members/joints, apply method of joints or a strategic section and state tension/compression sign convention.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not apply truss assumptions to a frame with moment-resisting joints or loads placed between joints without appropriate modelling.

2. Zero-force members and force paths−

Some members carry zero force for a given load case because of joint geometry. Recognising them can simplify analysis, but force state may change with another load case.

Study and application method

Inspect unloaded joints using the applicable collinearity rules, then confirm using equilibrium rather than deleting members blindly.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Inspect unloaded joints using the applicable collinearity rules, then confirm using equilibrium rather than deleting members blindly.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A zero-force result for one load case does not prove the member is unnecessary in the actual structure.

3. Slope and deflection−

Deflection is displacement under load; slope is rotation. Elastic-curve, moment-area, energy and virtual-work methods relate bending/strain energy to displacement.

Study and application method

State support conditions, loading, E and I/EA assumptions, choose an appropriate method and check whether direction/magnitude are physically reasonable.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State support conditions, loading, E and I/EA assumptions, choose an appropriate method and check whether direction/magnitude are physically reasonable.

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Common mistake / precaution: Serviceability deflection and vibration limits matter even when strength is adequate.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Indeterminacy, Stability and Application

Compatibility, stiffness/flexibility overview, continuous beams/frames, columns/buckling, influence of loads and responsible use of analysis tools.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

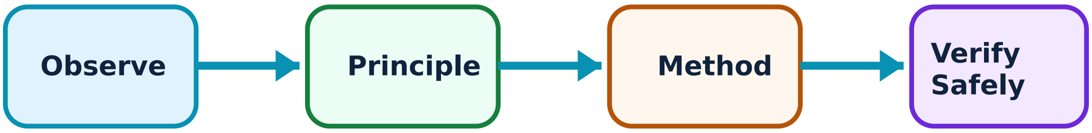
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Indeterminacy, Stability and Application: identify the system, state its principle, follow the correct method and verify the result safely.

1. Compatibility and indeterminate analysis−

Indeterminate structures have more unknowns than equilibrium equations. Deformations must be compatible with supports and connections, so stiffness/flexibility, slope-deflection or moment-distribution approaches are used.

Study and application method

Identify redundants, state compatibility condition and outline how member stiffness and joint rotation enter the chosen method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify redundants, state compatibility condition and outline how member stiffness and joint rotation enter the chosen method.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not treat software output as correct without validating model, support conditions, loads and units.

2. Columns and buckling−

Compression members can lose stability by buckling before material strength is reached. Effective length, end restraint, slenderness, stiffness and imperfections govern behaviour.

Study and application method

Sketch end conditions and likely buckling mode; distinguish short-column strength from slender-column instability and state data needed for check.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch end conditions and likely buckling mode; distinguish short-column strength from slender-column instability and state data needed for check.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Euler-style ideal formulae have limited applicability; real design needs code-based treatment of imperfections and restraint.

3. Model validation and communication−

A defensible analysis states assumptions, inputs, method, results and limitations. Checks include equilibrium, units, deformation shape, boundary conditions and comparison with simple cases.

Study and application method

Prepare a concise analysis sheet with model diagram, load table, sign convention, calculations/results and independent reasonableness checks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Never use an analysis result for construction without authorised design review.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Structural Modelling and Equilibrium

Use the concept flow to revise observation, principle, method and verification.

Module 2: Shear Force, Bending Moment and Stresses

Use the concept flow to revise observation, principle, method and verification.

Module 3: Trusses and Deflections

Use the concept flow to revise observation, principle, method and verification.

Module 4: Indeterminacy, Stability and Application

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Create free-body diagrams and calculate reactions for supplied beam load cases.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw and check SFD/BMD for point and distributed load examples.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Analyse a small pin-jointed truss by joints or sections and identify member force sense.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare determinate and indeterminate models of a beam/frame and discuss effects on deflection and redundancy.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Structural Modelling and Equilibrium

Explain Loads, supports and free-body diagrams and state one correct method, application or precaution.−

A structural model represents members, supports and applied loads with idealised symbols. Pinned, roller and fixed supports restrain different motions and therefore develop different reactions.

Answer framework: Isolate the body, replace supports with unknown reactions, show all loads/dimensions and state the coordinate/sign convention before writing equations.

Important point: Never calculate reactions from an incomplete free-body diagram or without units.

Explain Equilibrium and reactions and state one correct method, application or precaution.−

A planar structure in static equilibrium satisfies zero resultant force in two directions and zero resultant moment. These equations determine reactions for statically determinate systems.

Answer framework: Choose moment centre strategically, solve equations systematically and check force/moment balance after obtaining results.

Important point: Keep distributed-load resultants, locations and signs correct; a correct magnitude at a wrong location gives a wrong moment.

Explain Determinacy and stability and state one correct method, application or precaution.−

Determinate structures can be solved using equilibrium alone; indeterminate structures need compatibility and stiffness/deformation relations. A model must also be stable against rigid-body motion.

Answer framework: Count unknown reactions/member forces relative to independent equations and inspect whether the support/member arrangement prevents translation and rotation.

Important point: Do not confuse an over-restrained diagram with a stable, well-modelled real structure.

Module 2 — Shear Force, Bending Moment and Stresses

Explain Section-cut internal actions and state one correct method, application or precaution.−

A cut through a loaded member reveals internal axial force, shear force and bending moment needed to keep the isolated segment in equilibrium.

Answer framework: Find reactions first, cut at a location, choose one side, apply the declared sign convention and write equilibrium equations.

Important point: Use one sign convention consistently throughout diagrams and calculations.

Explain Shear-force and bending-moment diagrams and state one correct method, application or precaution.−

SFD/BMD show how internal shear and moment vary along a beam. Point loads create shear jumps; distributed loads change shear slope; extrema of moment occur where shear crosses zero.

Answer framework: Mark key positions, calculate values on either side of discontinuities, draw the correct line/curve shape and verify end/support conditions.

Important point: Do not draw diagrams from memorised shapes before computing reactions and load intervals.

Explain Bending and shear stress and state one correct method, application or precaution.−

Bending moment produces normal stress varying with distance from the neutral axis; transverse shear produces a non-uniform shear-stress distribution related to section geometry.

Answer framework: State assumptions, identify M/V and section properties, use consistent units and locate the fibre/point where stress is required.

Important point: Do not apply a beam formula outside its assumptions or mix units for force, length and section properties.

Module 3 — Trusses and Deflections

Explain Pin-jointed truss analysis and state one correct method, application or precaution.−

An ideal plane truss carries loads at joints and members act mainly in axial tension or compression. Method of joints uses joint equilibrium; method of sections isolates a portion efficiently.

Answer framework: Check global reactions, label members/joints, apply method of joints or a strategic section and state tension/compression sign convention.

Important point: Do not apply truss assumptions to a frame with moment-resisting joints or loads placed between joints without appropriate modelling.

Explain Zero-force members and force paths and state one correct method, application or precaution.−

Some members carry zero force for a given load case because of joint geometry. Recognising them can simplify analysis, but force state may change with another load case.

Answer framework: Inspect unloaded joints using the applicable collinearity rules, then confirm using equilibrium rather than deleting members blindly.

Important point: A zero-force result for one load case does not prove the member is unnecessary in the actual structure.

Explain Slope and deflection and state one correct method, application or precaution. −

Deflection is displacement under load; slope is rotation. Elastic-curve, moment-area, energy and virtual-work methods relate bending/strain energy to displacement.

Answer framework: State support conditions, loading, E and I/EA assumptions, choose an appropriate method and check whether direction/magnitude are physically reasonable.

Important point: Serviceability deflection and vibration limits matter even when strength is adequate.

Module 4 — Indeterminacy, Stability and Application

Explain Compatibility and indeterminate analysis and state one correct method, application or precaution. −

Indeterminate structures have more unknowns than equilibrium equations. Deformations must be compatible with supports and connections, so stiffness/flexibility, slope-deflection or moment-distribution approaches are used.

Answer framework: Identify redundants, state compatibility condition and outline how member stiffness and joint rotation enter the chosen method.

Important point: Do not treat software output as correct without validating model, support conditions, loads and units.

Explain Columns and buckling and state one correct method, application or precaution. −

Compression members can lose stability by buckling before material strength is reached. Effective length, end restraint, slenderness, stiffness and imperfections govern behaviour.

Answer framework: Sketch end conditions and likely buckling mode; distinguish short-column strength from slender-column instability and state data needed for check.

Important point: Euler-style ideal formulae have limited applicability; real design needs code-based treatment of imperfections and restraint.

Explain Model validation and communication and state one correct method,

application or precaution.−

A defensible analysis states assumptions, inputs, method, results and limitations. Checks include equilibrium, units, deformation shape, boundary conditions and comparison with simple cases.

Answer framework: Prepare a concise analysis sheet with model diagram, load table, sign convention, calculations/results and independent reasonableness checks.

Important point: Never use an analysis result for construction without authorised design review.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Structural Modelling and Equilibrium.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Shear Force, Bending Moment and Stresses.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Trusses and Deflections.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Indeterminacy, Stability and Application.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Types of structures, loads, supports, free-body diagrams, reactions, equilibrium, determinacy and sign conventions..
2. Develop a complete answer or practical plan for: Section cuts, shear/moment sign convention, SFD/BMD, load-shear-moment relations, bending and shear stress..
3. Develop a complete answer or practical plan for: Pin-jointed trusses, methods of joints/sections, zero-force members, strain energy/virtual work concepts, slope and beam

deflection..
4. Develop a complete answer or practical plan for: Compatibility, stiffness/flexibility overview, continuous beams/frames, columns/buckling, influence of loads and responsible use of analysis tools..

LAST-MILE REVISION

Quick Revision

Module 1: Structural Modelling and Equilibrium

Types of structures, loads, supports, free-body diagrams, reactions, equilibrium, determinacy and sign conventions.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Shear Force, Bending Moment and Stresses

Section cuts, shear/moment sign convention, SFD/BMD, load-shear-moment relations, bending and shear stress.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Trusses and Deflections

Pin-jointed trusses, methods of joints/sections, zero-force members, strain energy/virtual work concepts, slope and beam deflection.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Indeterminacy, Stability and Application

Compatibility, stiffness/flexibility overview, continuous beams/frames, columns/buckling, influence of loads and responsible use of analysis tools.

- Match each topic to CO4.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Loads, supports and free-body diagrams	A structural model represents members, supports and applied loads with idealised symbols.
Equilibrium and reactions	A planar structure in static equilibrium satisfies zero resultant force in two directions and zero resultant moment.
Determinacy and stability	Determinate structures can be solved using equilibrium alone; indeterminate structures need compatibility and stiffness/deformation relations.
Section-cut internal actions	A cut through a loaded member reveals internal axial force, shear force and bending moment needed to keep the isolated segment in equilibrium.
Shear-force and bending-moment diagrams	SFD/BMD show how internal shear and moment vary along a beam.
Bending and shear stress	Bending moment produces normal stress varying with distance from the neutral axis; transverse shear produces a non-uniform shear-stress distribution related to section geometry.
Pin-jointed truss analysis	An ideal plane truss carries loads at joints and members act mainly in axial tension or compression.
Zero-force members and force paths	Some members carry zero force for a given load case because of joint geometry.
Slope and deflection	Deflection is displacement under load; slope is rotation.
Compatibility and indeterminate analysis	Indeterminate structures have more unknowns than equilibrium equations.
Columns and buckling	Compression members can lose stability by buckling before material strength is reached.
Model validation and communication	A defensible analysis states assumptions, inputs, method, results and limitations.

LEARNING RESOURCES

References

Recommended theory-of-structures references

1. R. C. Hibbeler, Structural Analysis.
2. S. S. Bhavikatti, Structural Analysis.
3. R. K. Rajput, Strength of Materials.

Approved public structural-analysis sources

- https://mechanicalc.com/reference/beam-analysis
- https://testbook.com/civil-engineering/theory-of-structures

These sources support the study handbook while official Course 4012 content is unavailable; they do not replace official syllabus requirements.